

Exercise 2.1: Convert the following linear program to standard form:

$$\begin{aligned} \text{Maximize: } z &= 3x_1 + 5x_2 - 4x_3 \\ \text{Subject to: } 7x_1 - 2x_2 - 3x_3 &\geq 4 \\ -2x_1 + 4x_2 + 8x_3 &= -3 \\ 5x_1 - 3x_2 - 2x_3 &\leq 9 \\ x_1 &\geq 1, x_2 \leq 7, x_3 \geq 0 \end{aligned}$$

Solution. Converting this LP problem to standard form we must multiply our $\hat{c}$ by -1 to turn this into a minimization problem. Then we will add an excess variable to the $7x_1 - 2x_2 - 3x_3 \geq 4$ constraint, we must then invert the $-2x_1 + 4x_2 + 8x_3 = -3$ to make ensure that our $\hat{b} \geq 0$. Then we will add a slack variable on $5x_1 - 3x_2 - 2x_3 \leq 9$. Finally we need to ensure that all of our individual inequality constraints are positivity constraints, so $x_1 \geq 1$ will be replace by a variable $x'_1 = x_1 - 1$ and $x'_1 \geq 0$. We will also replace $x_2 \leq 7$ with $x'_2 = 7 - x_2$ and $x'_2 \geq 0$. Finally we will have to substitute our new variables into the entire problem. Doing all that we get that the following LP problem in standard form,

$$\begin{aligned} \text{Minimize: } z &= -3x'_1 + 5x'_2 + 4x_3 + 0e_1 + 0s_1 - 38 \\ \text{Subject to: } 7x'_1 + 2x_2 - 3x_3 - e_1 &= 11 \\ 2x'_1 + 4x'_2 - 8x_3 &= 29 \\ 5x'_1 + 3x'_2 - 2x_3 + s_1 &= 25 \\ x'_1, x'_2, x_3, e_1, s_1 &\geq 0 \end{aligned}$$

□

Exercise 2.2: Convert the following linear program to standard form:

$$\begin{aligned} \text{Minimize: } z &= x_1 - 5x_2 - 7x_3 \\ \text{Subject to: } 5x_1 - 2x_2 + 6x_3 &\geq 5 \\ 3x_1 + 4x_2 - 9x_3 &= 3 \\ 7x_1 + 3x_2 + 5x_3 &\leq 9 \\ x_1 &\geq -2, x_2 \text{ and } x_3 \text{ free.} \end{aligned}$$

Solution. Converting this LP problem to standard form must add an excess variable to the $5x_1 - 2x_2 + 6x_3 \geq 5$ constraint, and add a slack variable to the $7x_1 + 3x_2 + 5x_3 \leq 9$ constraint. Note that $x_1 \geq -2$ will be replaced by a variable $x'_1 = x_1 + 2$ and $x'_1 \geq 0$. Also since we have free variables we will be replacing them with $x_2 = x_4 - x_5$ and $x_3 = x_6 - x_7$ with

$x_4, x_5, x_6, x_7 \geq 0$. Doing all that we get the following LP problem in standard form,

$$\begin{aligned} \text{Minimize: } z &= x'_1 - 5x_4 + 5x_5 - 7x_6 + 7x_7 + 0e_1 + 0s_1 - 2 \\ \text{Subject to: } 5x'_1 - 2x_4 + 2x_5 + 6x_6 - 6x_7 - e_1 &= 5 \\ 3x'_1 + 4x_4 - 4x_5 - 9x_6 + 9x_7 &= 9 \\ 7x'_1 + 3x_4 - 3x_5 + 5x_6 - 5x_7 + s_1 &= 23 \\ x'_1, x_4, x_5, x_6, x_7, e_1, s_1 &\geq 0. \end{aligned}$$

□

Exercise 2.4: Consider the previous LP problem. Convert it to standard form without the $x_3 = x_7 - x_7$ substitution. Show that the problem can be replaced by an equivalent problem with one less variable and one less constraint by eliminating x_3 using the equality constraints. Why cannot this technique be used to eliminate variables with nonnegativity constraints?

Solution. Converting to standard form without the $x_3 = x_7 - x_7$ substitution gives us the following LP problem,

$$\begin{aligned} \text{Minimize: } z &= x'_1 - 5x_4 + 5x_5 - 7x_3 + 0e_1 + 0s_1 - 2 \\ \text{Subject to: } 5x'_1 - 2x_4 + 2x_5 + 6x_3 - e_1 &= 5 \\ 3x'_1 + 4x_4 - 4x_5 - 9x_3 &= 9 \\ 7x'_1 + 3x_4 - 3x_5 + 5x_3 + s_1 &= 23 \\ x'_1, x_4, x_5, e_1, s_1 &\geq 0. \text{ With } x_3 \text{ free.} \end{aligned}$$

Performing the following eliminations: Taking the first inequality and adding $\frac{2}{3}$ of the second,

$$\begin{array}{r} 5x'_1 - 2x_4 + 2x_5 + 6x_3 - e_1 = 5 \\ \frac{2}{3}(3x'_1 + 4x_4 - 4x_5 - 9x_3 = 9) + \\ \hline 7x'_1 + \frac{2}{3}x_4 - \frac{2}{3}x_5 - e_1 = 11 \end{array}$$

Tanking the third inequality and adding $\frac{5}{9}$ of the second,

$$\begin{array}{r} 7x'_1 + 3x_4 - 3x_5 + 5x_3 + s_1 = 23 \\ \frac{5}{9}(3x'_1 + 4x_4 - 4x_5 - 9x_3 = 9) + \\ \hline \end{array}$$

$$\frac{16}{3}x'_1 + \frac{47}{9}x_4 - \frac{47}{9}x_5 + s_1 = 32$$

Taking the cost function and adding $-\frac{7}{9}$ of the second,

$$\begin{array}{r} x'_1 - 5x_4 + 5x_5 - 7x_3 + 0e_1 + 0s_1 = z + 2 \\ -\frac{7}{9}(3x'_1 + 4x_4 - 4x_5 - 9x_3 = 9) + \\ \hline \end{array}$$

$$-\frac{12}{9}x'_1 - \frac{73}{9}x_4 + \frac{23}{9}x_5 + 0e_1 + 0s_1 + 5 = z$$

Putting everything together we get the following LP problem with one less variable and one less constraint,

$$\begin{array}{l} \text{Minimize: } z = -\frac{12}{9}x'_1 - \frac{73}{9}x_4 + \frac{23}{9}x_5 + 0e_1 + 0s_1 + 5 \\ \text{Subject to: } 7x'_1 + \frac{2}{3}x_4 - \frac{2}{3}x_5 - e_1 = 11 \\ \frac{16}{3}x'_1 + \frac{47}{9}x_4 - \frac{47}{9}x_5 + s_1 = 32 \\ x'_1, x_4, x_5, e_1, s_1 \geq 0. \end{array}$$

As outlined in the example in the book (p.104 Figure 4.2) performing elimination on inequality constraints produces a new system with a different feasible region. (Also this would've been cleaner in matrix form) $\square$

Exercise 3.1: Consider the system of linear constraints

$$\begin{array}{l} 3x_1 + x_2 \leq 100 \\ x_1 + x_2 \leq 80 \\ x_1 \leq 40 \\ x_1, x_2 \geq 0 \end{array}$$

- (i) Write this system of constraints in standard form, and determine all the basic solutions (feasible and infeasible).

Solution. Writing this system of constraints in standard form we get the following,

$$2x_1 + x_2 + s_1 = 100$$

$$x_1 + x_2 + s_2 = 80$$

$$x_1 + s_3 = 40$$

$$x_1, x_2 \geq 0$$

Converting this into a system we get,

$$\begin{pmatrix} 2 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{pmatrix} x = \begin{pmatrix} 100 \\ 80 \\ 40 \end{pmatrix}.$$

Simplifying to RREF we get,

$$\begin{pmatrix} 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 & -1 \\ 0 & 0 & 1 & -1 & -1 \end{pmatrix} x = \begin{pmatrix} 40 \\ 40 \\ -20 \end{pmatrix}.$$

Therefore solutions to this equation with respect to our free slack variables are of the form,

$$x = \begin{pmatrix} 40 \\ 40 \\ -20 \\ 0 \\ 0 \end{pmatrix} + s_2 \begin{pmatrix} 0 \\ -1 \\ 1 \\ 1 \\ 0 \end{pmatrix} + s_3 \begin{pmatrix} -1 \\ 1 \\ 1 \\ 0 \\ 1 \end{pmatrix}$$

We say that non-negative solutions are feasible and negative solutions are infeasible.

□

- (ii) Determine the extreme points of the feasible region (corresponding to both the standard form of the constraints, as well as the original version).

Solution.

□

Exercise 3.12: Consider the system of constraints $Ax = b$, $x \geq 0$ with

$$A = \begin{pmatrix} 1 & 4 & 7 & 1 & 0 & 0 \\ 2 & 5 & 8 & 0 & 1 & 0 \\ 3 & 6 & 9 & 0 & 0 & 1 \end{pmatrix} \text{ and } b = \begin{pmatrix} 12 \\ 15 \\ 18 \end{pmatrix}. \quad (1)$$

Is $x = (1, 1, 1, 0, 0, 0)^T$ a basic feasible solution? Explain your answer.

Solution. First we need to check if x does satisfy the constraints,

$$1 + 4 + 7 = 12$$

$$2 + 5 + 8 = 15$$

$$3 + 6 + 9 = 18$$

and since $x \geq 0$ we know that this solution is a feasible one. To determine if the solution is basic we need to find out if the nonzero entries of x form a basis in the row space. A quick computation of the determinant tells us that the columns are not linearly independent,

$$\det \begin{pmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{pmatrix} = 1(-3) - 4(-6) + 7(-3) = 0.$$

So x is only a feasible solution, not a basic feasible solution. $\square$

Exercise 3.16: Let $S = \{x : x^T x \leq 1\}$. Prove that the extreme points of S are the points on its boundary.

Proof. Suppose $S = \{x : x^T x \leq 1\}$ and let x be an extreme point of S . By definition an extreme point cannot be written as a convex combination of two interior points. Let $a, b \in S$ such that a and b are interior points. By definition we know that $a^T a < 1$ and $b^T b < 1$. Now consider the convex combination of these points, for some $0 < \sigma < 1$ we have, $\sigma(a) + (1 - \sigma)(b)$. Now consider the inner product of this vector

$$(\sigma(a) + (1 - \sigma)(b))^T (\sigma(a) + (1 - \sigma)(b)) = \sigma^2(a^T a) + (1 - \sigma)^2(b^T b) + 2\sigma(1 - \sigma)(a^T b) \quad (2)$$

Since $(a^T a)$ and $(b^T b)$ are less than 1, we also know that $(a^T b)$ is less than 1 and by substitution we get that,

$$(\sigma(a) + (1 - \sigma)(b))^T (\sigma(a) + (1 - \sigma)(b)) < \sigma^2 + (1 - \sigma)^2 + 2\sigma(1 - \sigma) = 1 \quad (3)$$

Therefore every boundary point is an extreme point. Clearly if a point is not on the boundary then it is in the interior and therefore not extreme. Thus every extreme points of S is on its boundary. $\square$

Exercise Problem P9: Write a Matlab function which uses QR decomposition to construct an orthonormal null space matrix for a given full row rank $m \times n$ matrix.

Solution. Consider the following Matlab function,

Code:

```
function Z = mynull(A)
    m = size(A, 1)
    n = size(A, 2)
    A = A'
    [Q, R] = qr(A)
    Z = Q(:, (m+1):n)
end
```

Console:

```
-----
>> A = [1 2 3 4 5; 6 7 8 9 10; 4 1 0 1 4]
```

```
A =
     1     2     3     4     5
     6     7     8     9    10
     4     1     0     1     4
```

```
>> Z = mynull(A)
```

```
Z =
    -0.0222    -0.3373
     0.2731     0.7443
    -0.6860    -0.2090
     0.6416    -0.4657
    -0.2065     0.2677
```

```
>> norm(A*Z - 0)
```

```
9.7664e-16
```

```
-----
>> A = [0 1 2 0]
```

```
A =
     0     1     2     0
```

```
>> Z = mynull(A)
```

```
Z =
    -0.4472    -0.8944         0
     0.8000    -0.4000         0
    -0.4000     0.2000         0
         0         0     1.0000
```

```
>> norm(A*Z - 0)
```

```
1.5701e-16
```

```
-----
>> A = [1 2; 3 4]
```

```
A =
```

$$\begin{matrix} 1 & 2 \\ 3 & 4 \end{matrix}$$

```
>> Z = mynull(A)
```

```
Z =
```

```
empty double matrix
```

```
>> norm(A*Z - 0)
```

```
0
```

Comparing both methods `mynull()` and `null()` do not differ at all. With every example we had `mynull(A) - null(A)` results in a matrix full of exact zeros (so the norm of this difference will also be 0). A quick dive into the documentation we find that the algorithm used for finding a null space is the SVD. $\square$